

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME:** Cryptography and Network Security  
**COURSE CODE:** CSA51

### COURSE OUTCOME COVERED

**CO1:** Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

**QUESTIONS: 5**

**Total Marks:50**

Annexure A

### CONCEPT MAPPING

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#### **Code of Conduct:**

*I Dhaanush Rajan - 192421162 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*



A handwritten signature in blue ink, appearing to read 'R. Dhaanush', is written over a faint, light blue rectangular stamp.

**Signature of the student:**

## CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
  - a. Show how each attack leads to specific threats and risks
  - b. Map how security goals help mitigate them Provide one real-world example for each link

### ANSWER

1. Malware Attack → Threat → Risk → Confidentiality

A **malware attack** occurs when malicious software such as a virus, worm, or ransomware infects a computer system. This creates the **threat** of unauthorized access to sensitive data or system compromise. As a result, organizations face **risks** such as data loss, financial loss, and privacy breaches. The security goal of **Confidentiality** helps mitigate this risk by using encryption, strong access controls, and authentication mechanisms to prevent unauthorized users from accessing confidential information. **Example:** The **WannaCry ransomware attack (2017)** encrypted files in hospitals and businesses worldwide, causing major financial and operational losses.

2. Phishing Attack → Threat → Risk → Confidentiality

A **phishing attack** involves attackers sending fake emails or websites to trick users into revealing their usernames, passwords, or banking details. This leads to the **threat** of credential theft. The resulting **risk** includes identity theft, unauthorized account access, and financial fraud. The security goal of **Confidentiality** reduces this risk by implementing multi-factor authentication (MFA), secure login procedures, and user awareness training. **Example:** A fake banking email asks customers to verify their account details, allowing attackers to steal login credentials.

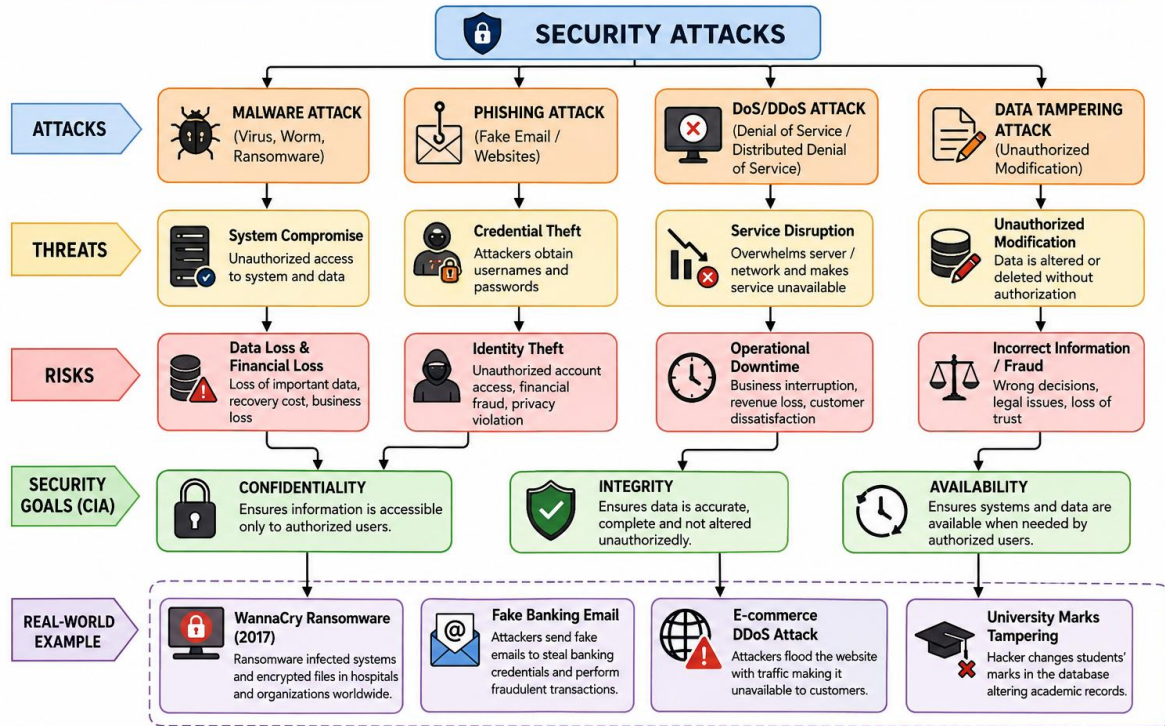
3. Denial-of-Service (DoS/DDoS) Attack → Threat → Risk → Availability

A **Denial-of-Service (DoS)** or **Distributed Denial-of-Service (DDoS)** attack floods a server or network with excessive traffic, preventing legitimate users from accessing services. This creates the **threat** of service disruption. The **risk** is operational downtime, customer dissatisfaction, and financial loss. The security goal of **Availability** helps mitigate this problem by using firewalls, load balancing, redundant servers, and DDoS protection services to ensure continuous system operation. **Example:** A DDoS attack on an e-commerce website makes the online shopping platform unavailable during a major sale.

4. Data Tampering Attack → Threat → Risk → Integrity

A **data tampering attack** occurs when an attacker modifies or deletes information without authorization. This results in the **threat** of corrupted or inaccurate data. The **risk** includes incorrect business decisions, fraud, and loss of customer trust. The security goal of **Integrity** ensures that data remains accurate and unaltered by using hashing, digital signatures, checksums, and strict access controls. **Example:** A hacker gains access to a university database and changes students' examination marks, resulting in inaccurate academic records.

# CONCEPT MAP: SECURITY ATTACKS, THREATS, RISKS AND SECURITY GOALS



2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

ANSWER

### 1. Caesar Cipher:

The Caesar Cipher is the simplest substitution cipher. It uses a single **shift key (0–25)** to replace every letter in the plaintext with another letter that is a fixed number of positions away in the alphabet. Since the same shift is applied to all characters, its complexity is **very low**, making it easy to implement. However, its security is **very weak** because there are only 26 possible keys, allowing attackers to easily crack it using brute-force or frequency analysis.

### 2. Monoalphabetic Cipher:

The Monoalphabetic Cipher replaces each plaintext letter with a unique ciphertext letter using a fixed substitution alphabet. Unlike the Caesar Cipher, it uses a much larger key space because the entire alphabet is rearranged. Its complexity is **medium**, as creating and managing the substitution table is more difficult. Although it provides better security than the Caesar Cipher, it is still vulnerable to frequency analysis since each letter is always replaced by the same ciphertext letter.

### 3. Vigenère Cipher:

The Vigenère Cipher is a polyalphabetic substitution cipher that uses a **repeating keyword** to determine different shifts for different letters in the plaintext. This makes the encryption process more complex and increases security because the same plaintext letter can be encrypted into different ciphertext letters. Its complexity is **high**, and its security is **strong** because it resists simple frequency analysis. However, if the keyword is discovered using cryptanalysis techniques such as the Kasiski examination, the cipher can still be broken.

### 4. Hill Cipher:

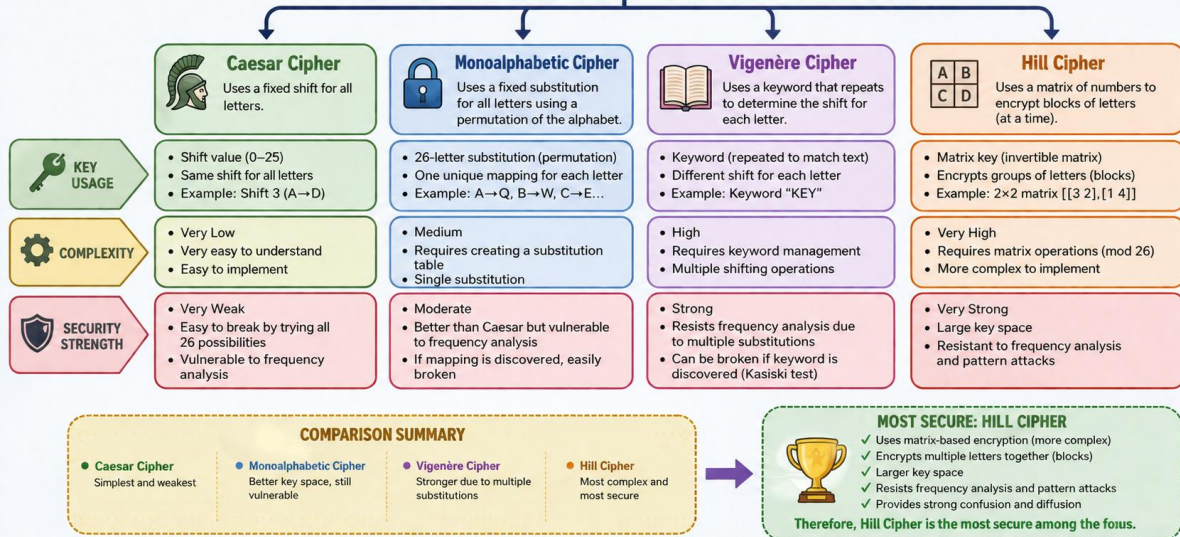
The Hill Cipher uses an **invertible matrix key** to encrypt blocks of letters instead of encrypting one character at a time. It relies on matrix multiplication and modular arithmetic, making its complexity **very high**. This block-based encryption hides letter frequency patterns and significantly increases the key space. As a result, the Hill Cipher provides **very strong security** and is much more resistant to frequency analysis and pattern-based attacks than the other substitution ciphers.

### 5. Comparison and Analysis:

Comparing all four techniques, the **Caesar Cipher** has the lowest complexity and weakest security, while the **Monoalphabetic Cipher** offers moderate security with a larger key space. The **Vigenère Cipher** improves security by using multiple substitution alphabets controlled by a keyword. Among all four techniques, the **Hill Cipher is the most secure** because it encrypts blocks of characters using matrix operations, provides better confusion and diffusion, has a larger key space, and effectively resists frequency analysis and many classical cryptographic attacks.

# SUBSTITUTION CIPHERS

Comparing Key Usage, Complexity and Security Strength



3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

- a. Show differences and similarities
  - i. Map how each affects the Data structure and Security level
  - ii. Analyze which is more vulnerable to attacks and why

ANSWER

Here is a detailed breakdown of the concept map for classical encryption schemes, expanding on the similarities, differences, data structure impacts, and vulnerability analyses.

## 1. Core Definitions & Shared Goals (Similarities)

At the top level, both **Substitution** and **Transposition** are the foundational pillars of classical cryptography.

**The Similarity:** Both techniques share the exact same core objective—to achieve **confusion/diffusion** and obscure the relationship between the plaintext and the ciphertext so an unauthorized party cannot read it.

**The Fundamental Difference:** They achieve this goal through entirely opposite mechanisms. Substitution changes the **identity** of the characters while keeping their positions, whereas Transposition changes the **position** of the characters while keeping their identities.

## 2. Deep Dive: Substitution vs. Transposition

### Substitution Techniques

**How it works:** Plaintext characters (or bits) are replaced by other characters, symbols, or numbers based on a defined key or rule.

#### Types:

*Monoalphabetic:* A single character always maps to the same ciphertext character (e.g., Caesar Cipher).

*Polyalphabetic:* A character maps to different ciphertext characters depending on its position in the text (e.g., Vigenère Cipher).

*Polygraphic:* Groups of characters are substituted together (e.g., Playfair Cipher).

**Effect on Data Structure:** It completely alters the **value/identity** of individual data units, but the underlying structural layout (the sequence flow) remains intact. For example, the word "CAT" might become "FDW".

**Security Level Profile:** Highly vulnerable to **Frequency Analysis** if it's monoalphabetic, because the unique statistical signature of a language (like the fact that 'E' and 'T' are the most common letters in English) passes right through the cipher. Polyalphabetic schemes flatten this frequency curve, raising the security level significantly.

### Transposition Techniques

**How it works:** Plaintext characters are not changed at all. Instead, their original order is shuffled, rearranged, or permuted using a geometric or algorithmic pattern.

## Types:

*Rail Fence:* Text is written diagonally across multiple "rails" and read off row by row.

*Columnar Transposition:* Text is written in a grid of fixed width and read column by column based on a keyword.

**Effect on Data Structure:** It preserves the exact **identity and values** of the original units but completely breaks apart the linear structure and adjacency of the data. For example, the word "ATTACK" might become "AKATTC".

**Security Level Profile:** Vulnerable to **Re-ordering and Anagramming attacks**. Because the exact letters of the plaintext are still present, a standard letter frequency count looks exactly like normal English text. This immediately tips off an attacker that a transposition cipher was used.

### 3. Vulnerability Analysis: Which is weaker and why?

**Verdict: Monoalphabetic Substitution** is generally the most vulnerable to simple, low-effort attacks, though **Transposition** becomes highly vulnerable if the message is long.

Why Monoalphabetic Substitution Fails Quickly:

1.

**Direct Mapping:** Every time an attacker cracks a single letter (e.g., figuring out that 'X' equals 'E'), they instantly decrypt that letter across the *entire* document.

**Frequency Fingerprint:** The unique statistical "fingerprint" of the language is preserved perfectly. A cipher text can often be broken in seconds using standard frequency charts.

Why Transposition is Moderately Stronger (but still flawed):

**Combinatorial Complexity:** To break a transposition cipher, an attacker has to guess the correct geometric rearrangement. For longer keys, the number of possible permutations grows rapidly.

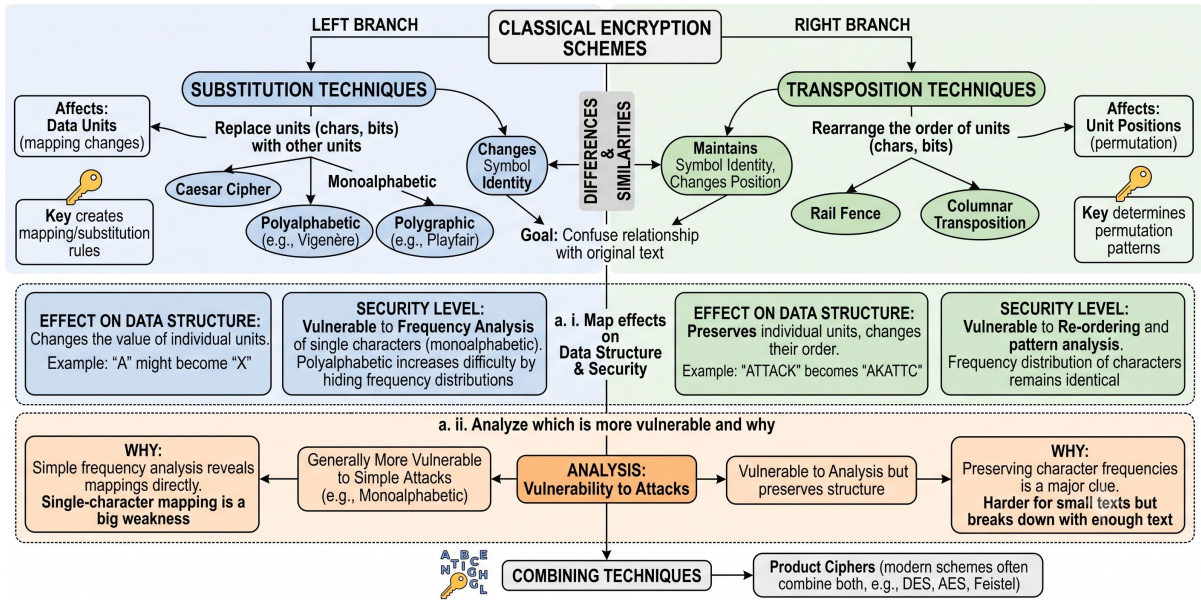
**The Catch:** Its biggest weakness is that it leaves a massive clue behind. Because the character frequencies match standard language frequencies perfectly, an attacker *knows* they just need to unscramble the letters rather than figure out a substitution code.

### 4. The Modern Resolution: Product Ciphers

Because both techniques have clear weaknesses when used alone, modern cryptography rarely relies on just one.

As shown at the bottom of the map, modern symmetric encryption standards (like **AES** and **DES**) combine *both* techniques into multiple iterations called **rounds**. By alternating substitution (for confusion) and transposition/permutation (for diffusion), they create incredibly secure **Product Ciphers** that classical frequency and anagramming attacks cannot break.

# CONCEPT MAP: CLASSICAL ENCRYPTION SCHEMES





4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
  - a. Show relationships between these concepts
  - b. Analyze how each contributes to secure communication
  - c. Include one real-world application for each

ANSWER

### 1. Data Hiding:

Data hiding is the primary concept of steganography, where secret information is embedded inside digital media such as images, audio, videos, or text without noticeably changing the original file. Its main contribution to secure communication is **confidentiality**, as the hidden message remains invisible to unauthorized users. A real-world application is **secure military or intelligence communication**, where confidential messages are concealed inside digital images before transmission.

### 2. Watermarking:

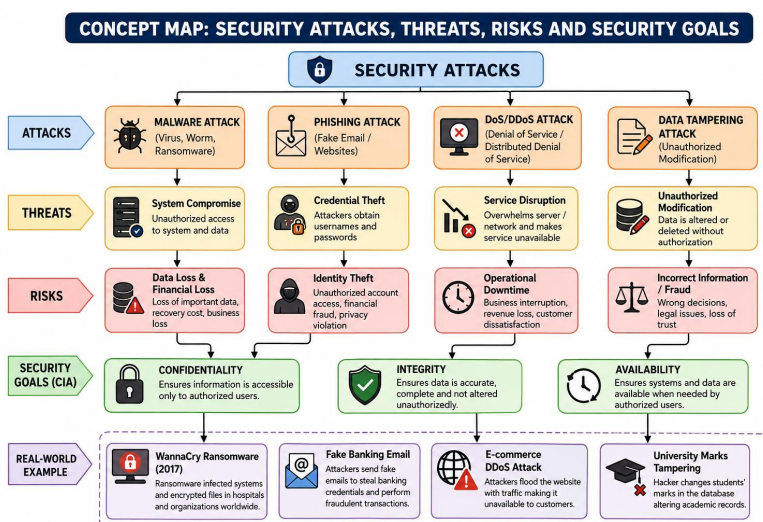
Watermarking involves embedding ownership, copyright, or authentication information into digital content. Unlike data hiding, its purpose is not secrecy but protecting intellectual property and verifying authenticity. It contributes to secure communication by ensuring that digital media can be authenticated and ownership can be proven even after distribution. A common real-world application is **copyright protection for digital images, videos, music, and online media**.

### 3. Survivability:

Survivability refers to the ability of hidden information or watermarks to remain intact even after common modifications such as image compression, resizing, cropping, filtering, or noise addition. It contributes to secure communication by ensuring the hidden information remains recoverable despite processing or attacks. A real-world application is **medical image transmission**, where patient information embedded in medical images remains available even after image compression during storage or transmission.

### 4. Relationship and Analysis:

These three concepts work together to achieve secure communication. **Data hiding** ensures confidentiality by concealing secret information, **watermarking** provides authentication and copyright protection, and **survivability** ensures that the embedded information remains reliable after modifications or attacks. Together, they strengthen digital security by protecting information confidentiality, verifying ownership, and maintaining data integrity in real-world communication systems.



5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
- Show how each concept is mathematically connected
  - Map their role in encryption/decryption
  - Analyze which concept is most critical for public key cryptography and justify

ANSWER

## 1. Modular Inverse

The **Modular Inverse** is a fundamental concept in number theory that finds a value  $a^{-1} \pmod{n}$  such that  $a \times a^{-1} \equiv 1 \pmod{n}$ . In RSA cryptography, it is used to calculate the **private key (d)** from the public key. Since the private key is essential for decryption, the modular inverse plays a critical role in ensuring secure communication.

## 2. Extended Euclidean Algorithm

The **Extended Euclidean Algorithm** is used to compute the **greatest common divisor (GCD)** of two numbers and to find the modular inverse when the numbers are relatively prime. In RSA, this algorithm calculates the private key **d** from the public key **e** and Euler's Phi Function. Without this algorithm, secure RSA key generation would not be possible.

## 3. Fermat's Little Theorem

**Fermat's Little Theorem** states that for a prime number  $p$ ,  $a^{p-1} \equiv 1 \pmod{p}$ . It is an important theorem in modular arithmetic and serves as a special case of Euler's Theorem. This theorem helps simplify modular exponentiation and provides the mathematical foundation for many cryptographic algorithms involving prime numbers.

## 4. Euler's Phi Function

The **Euler's Phi Function**, denoted as  $\phi(n)$ , calculates the number of positive integers less than  $n$  that are relatively prime to  $n$ . In RSA, it is computed using the prime factors of  $n$  and is used to generate both the public and private keys. Since determining  $\phi(n)$  for a large composite number is computationally difficult, it provides the core security of the RSA algorithm.

## 5. Euler's Theorem

**Euler's Theorem** states that  $a^{\phi(n)} \equiv 1 \pmod{n}$  when  $a$  and  $n$  are coprime. It extends Fermat's Little Theorem to composite numbers and provides the mathematical proof that RSA encryption and decryption correctly recover the original message. This theorem is one of the most important mathematical foundations of public-key cryptography.

## 6. Fundamental Proofs

The **Fundamental Proofs** combine modular arithmetic, Euler's Theorem, Fermat's Little Theorem, and the Euler's Phi Function to prove the correctness of RSA encryption and decryption. These mathematical proofs ensure that a message encrypted with the public key

can only be decrypted using the corresponding private key, guaranteeing secure communication.

## 7. RSA Key Generation

The **RSA Key Generation** process begins by selecting two large prime numbers and computing their product  $n$  along with Euler's Phi Function  $\phi(n)$ . The Extended Euclidean Algorithm is then used to calculate the private key from the public key. This process creates a pair of keys that enable secure public-key cryptography.

## 8. RSA Encryption

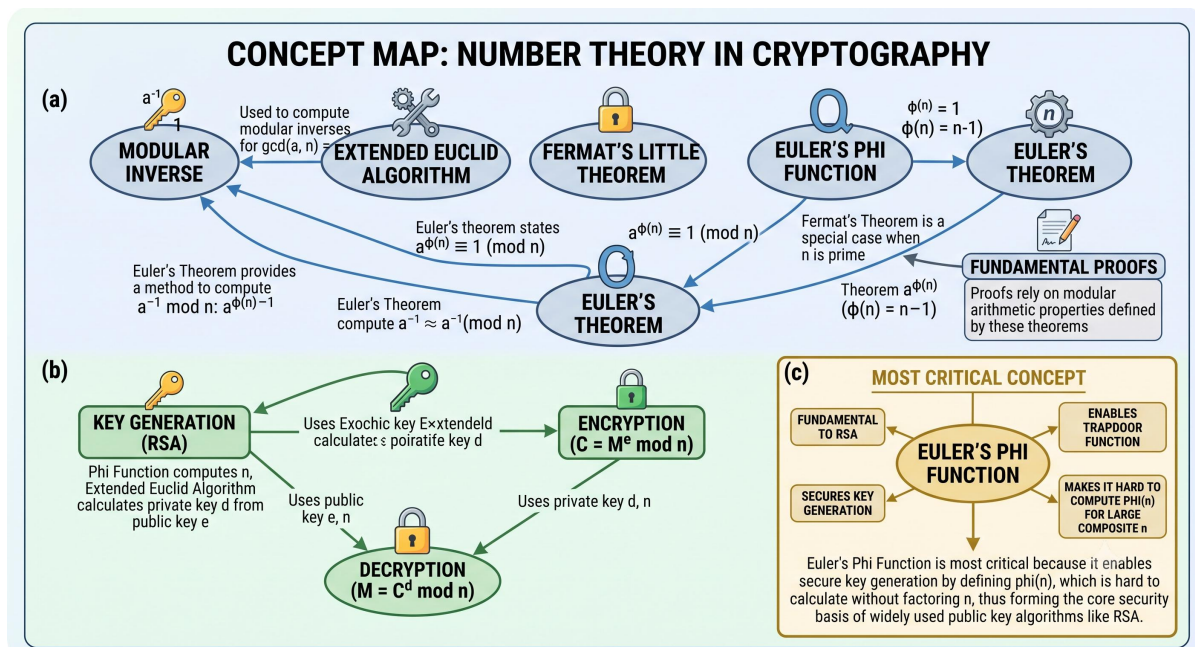
During **Encryption**, the sender uses the **public key**  $(e, n)$  to convert plaintext into ciphertext using the formula  $C = M^e \bmod n$ . This ensures that the message remains confidential because only the intended recipient possessing the private key can decrypt it.

## 9. RSA Decryption

During **Decryption**, the receiver uses the **private key**  $(d, n)$  to recover the original plaintext from the ciphertext using the formula  $M = C^d \bmod n$ . The correctness of this process is guaranteed by Euler's Theorem, ensuring secure and accurate communication.

## 10. Most Critical Concept – Euler's Phi Function

The concept map identifies **Euler's Phi Function** as the most critical concept because it forms the mathematical foundation of RSA cryptography. It enables secure key generation, supports the trapdoor function used in public-key encryption, and makes RSA secure by relying on the computational difficulty of factoring large composite numbers. Without Euler's Phi Function, RSA encryption and decryption would not be mathematically possible.



**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>           | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                                    | <b>Level 3 – Proficient</b>                                 | <b>Level 2 – Developing</b>                         | <b>Level 1 – Beginning</b>                       |
|---------------------------|------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25               | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30               | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20               | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |
| Analysis of Results       | 15               | Insightful analysis with accurate interpretation and validation of results                    | Good analysis with reasonable interpretation                | Basic analysis with limited insights                | Poor or incorrect analysis of results            |
| Documentation             | 10               | Clear, well-structured documentation with diagrams, code, and results                         | Organized documentation with necessary details              | Basic documentation with missing details            | Poor or incomplete documentation                 |